

Theoretical Physics & Mathematical Equations

Mathematical expressions and LaTeX formulas validation.

1. Mass-Energy Equivalence Formula:

$$E = m c^2$$

2. Gaussian Normal Distribution Density:

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

3. Standard Deviation Summation:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$